

Vaccuperm VGA-117

Gas dosing regulator

GB Installation and operating instructions

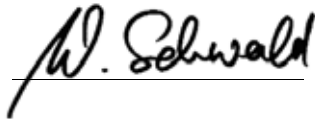


Declaration of conformity

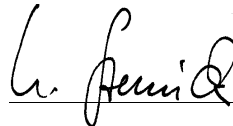
We, **Grundfos Alldos**, hereby declare under our sole responsibility that the products **Vaccuperm VGA-117-390 and -395 with electrical equipment (with CE mark on nameplate)**, to which this declaration relates, comply with the following EC Directives for harmonisation with the legal requirements of the EC member states:

- Electrical equipment designed for use within certain voltage limits (73/23/EEC).
Standard which has been used:
EN 60204-1:2006
- Electromagnetic compatibility (89/336/EEC).
Standards which have been used:
EN 61000-6-1:2007
EN 61000-6-3:2007
- Other bodies of rules applied:
DIN 19606:2006

26th January 2009



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Warning

These operating instructions are also available on www.Grundfosalldos.com.

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.



1. Of general interest

1.1 Structure of the documentation

The Grundfos Alldos dosing regulator VGA-117 is a state-of-the-art solution, which complies with recognised safety regulations. Conformity with applicable standards, directives and laws has been verified.

Nevertheless, certain risks which cannot be prevented by the manufacturer are associated with the use of the system.

Purpose of this manual:

- Inform users of optimum use.
- Warn users of possible residual risks when using correctly and identify measures that should be taken to avoid damage.
- Caution users against obvious misuse or inappropriate use and inform them of the necessary care that must be taken when operating the system.

1.2 About this manual

This manual contains the following standardised safety instructions about possible residual risks:



Warning

If these safety instructions are not observed, it may result in personal injury!

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!

Note

Notes or instructions that make the job easier and ensure safe operation.

Information about possible residual risks is provided:

- On warning signs displayed in the installation location.
- At the start of each section in this manual.
- Directly before any operating procedures that could involve residual risks.

1.3 User/target groups

Users are persons who are responsible for operating and monitoring the dosing regulator VGA-117 at the installation location. The system may only be operated by trained and qualified personnel. Personnel must have appropriate technical knowledge and be familiar with the basic principles of measuring and control technology.

1.3.1 Responsibilities of the users

The users' responsibilities:

- Read this manual before operating the VGA-117.
- Be trained by qualified personnel from Grundfos Alldos in the operation of the system.
- Observe the recognised regulations governing safety in the workplace and accident prevention.
- Wear appropriate protective clothing in accordance with national regulations for the prevention of accidents when operating the system and handling chemicals (German GUV-V D05).

1.4 Responsibilities of the operator

The owner of the building or operator of the VGA-117 is responsible for the following:

- Consider this manual to be part of the product and ensure that it is kept clearly accessible in the immediate vicinity of the system for the entire service life of the system.
- Meet the installation requirements specified by the manufacturer (required water connections and fittings, environmental conditions, electrical connection, protective pipe for dosing line if necessary, audible or optical warning device for alarm messages if necessary).
- Ensure that water lines and fixings are regularly checked, serviced and maintained.
- Obtain official approval for storing chemicals, if necessary.
- Train users in the operation of the system.
- Ensure that the regulations for the prevention of accidents are observed in the installation location (German GUV-V D05 regulation for the prevention of accidents, "Chlorination of Water" dated January 1997).
- Provide all users and service personnel with protective clothing in accordance with GUV-V D05 (face mask, gloves, protective apron).

1.5 Maintenance and service personnel

The system may only be maintained and serviced by authorised service personnel from Grundfos Alldos.

1.6 Correct usage

Grundfos Alldos VGA-117 may be used for dosing chlorine Cl_2 , as being described in this manual.

1.7 Inappropriate usage

Applications other than those listed in section 1.6 *Correct usage* are considered not to be in accordance with the intended use and are not permitted. The manufacturer, Grundfos Alldos, accepts no liability for any damage resulting from incorrect use.

The system comprises state-of-the-art components and has undergone safety-related testing.

Warning

Unauthorised structural modifications to the system may result in serious damage to equipment and personal injury.

It is forbidden to open, modify, change the structure of, bridge, remove, bypass or disable components, especially safety equipment.



2. Handling chlorine

2.1 Physical and chemical data

Under normal conditions of pressure and temperature, chlorine is a yellowish green gas with a pungent odour. It exists as diatomic molecule Cl_2 .

It is not flammable, but can under certain circumstances promote the flammability of metals, hydrocarbons etc.

Atomic weight	35.457
Molecular weight Cl_2	70.941
Density (liquid)	1.57 g/cm ³ at -34.05 °C
Density (gaseous)	3,214 g/l at 0 °C, 1 bar
1 l liquid chlorine at 0 °C	corresponds to 457 l (0.457 m ³) gaseous chlorine
1 kg liquid chlorine at 0 °C	corresponds to 311 l (0.311 m ³) gaseous chlorine
Specific gravity	2.486 (specific gravity of air: 1)
Boiling point	- 34.05 °C (1 bar)
Melting point	- 100.98 °C
Evaporation heat	269 kJ/kg (at 0 °C)
Heat conductivity	0.527 kJ/m2h (liquid chlorine)
Degree of purity acc. to DIN 19607	99,5 %
TLV (Threshold Limit Value)	1.5 mg/m3 (0.5 Vol.-ppm)

Vapour pressure curve of chlorine

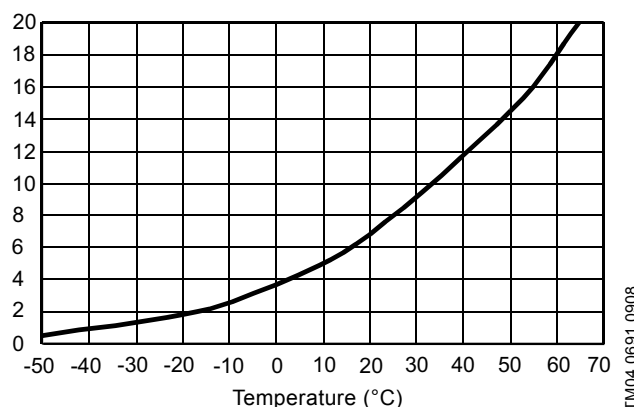


Fig. 1 Vapour pressure curve of chlorine

Solubility of chlorine gas in water

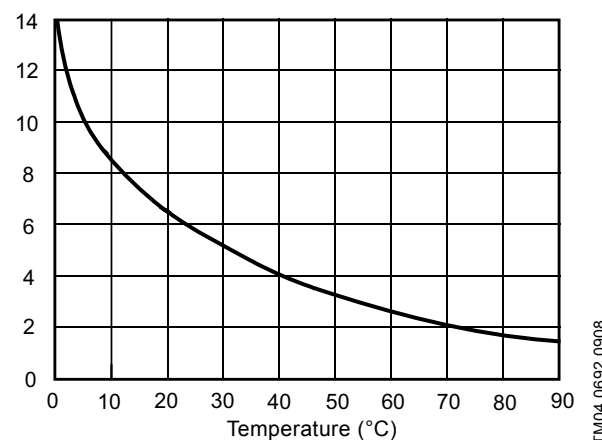


Fig. 2 Solubility of chlorine gas in water

2.2 Safety advice for handling chlorine

2.2.1 Risks to health

Chlorine gas is toxic, more than 50 Vol.-ppm in the indoor air mean an acute danger to life.



Warning

R 23

Toxic by inhalation.

R 36/37/38

Irritating to eyes, respiratory system and skin.

Hazards of chlorine gas

- Irritating to eyes, respiratory system and skin.
- Causes whooping cough.
- Causes causticization of skin and respiratory system.
- Lethal by lung edema in case of long influence or high concentration.
- Slight paralyzing effect to the central nervous system.

Hazards of liquid chlorine

- Causes causticization of skin.
- Causes reddening and formation of bubbles.

2.2.2 Personal safety equipment

The operating authority of a chlorination plant has to provide for the operation personnel:

For each person

Respiratory equipment (full-sight gas mask)

- with an effective chlorine filter (B2P3), colour mark: grey with a white ring
- at least 1 spare filter per gas mask
- personally fitted (perfectly tight)
- labelled by name

Only plants with chlorine drums

- at least 2 protective suits with compressed air respirators

Storage of the safety equipment

- outside the chlorine rooms
- well visible
- easily available at any time
- protected from dust and moisture

Caution *Further obligations of the operating authority*

- Introducing the operation personnel to handle the safety equipment
- Carrying out exercises (at least half-yearly)
- Regular replacement of the gas mask filters
 - after the expiry of the date of durability
 - at least 6 months after opening (mark the opening date on the filter)
 - after contact with chlorine
 - Observe employing prohibition according to § 14 ArbStoffV (in Germany) resp. according to local laws!

2.2.3 Rules of conduct

- Change of chlorine containers only with gas mask.
- Entering contaminated rooms only with protective suit and compressed air respirator.
- In case of flight wear gas mask, if possible. Observe wind direction!
- Eating, drinking and storing food is prohibited in chlorine rooms.

2.2.4 First aid in case of accidents

First aid after having inhaled chlorine

- Keep calm.
- Remove injured persons from the dangerous area.
 - Helpers must pay attention to personal protection!
- Immediately remove contaminated clothes.
- Calm down injured persons and keep them warm with blankets.
- Supply fresh air; use oxygen respirator (alternately with inhaling steam), if possible.
 - No mouth-to-mouth resuscitation!
- Fast and gentle transport to hospital
 - lying
 - sitting in case of difficulty in breathing
 - state chlorine causticization as the cause.

First aid after causticization of the skin

- Keep calm.
- Remove contaminated clothes.
- Rinse skin with plenty of water
- Bandage the wound germ-freely
- Seek medical aid.
 - State chlorine causticization as the cause.

First aid after causticization of the eyes

- Keep calm.
- Rinse causticized eyes with plenty of water while the person is lying.
 - Protect healthy eye, if necessary.
 - Spread eyelids widely, let the eye move to all sides.
- Seek ophthalmologist.
 - State chlorine causticization as the cause.

First aid after internal causticization

- Keep calm.
- Drink water in short sips.
 - If possible, take medical charcoal.
- Seek medical aid.
 - state chlorine causticization as the cause.

2.2.5 Transport and storage of chlorine



Warning

Handling of chlorine containers only by experienced, practised personnel!

Basic rules for transport and storage of chlorine

- Treat containers carefully, do not throw!
- Protect containers from turning over or rolling away!
- Protect containers from direct sun rays and temperatures over 50 °C!
- Transport of containers only with valve protection nut and protection cap.



Warning

This rules are valid for both full and empty containers, as empty containers still contain rests of chlorine and therefore are under pressure.

Valid regulations

- Regulations for accident prevention "Chlorination of water" (GUV-V D5) with process instructions
- Regulations concerning places of work (ArbStättV))
- Technical rules for gases 280, 310 und 330



Warning

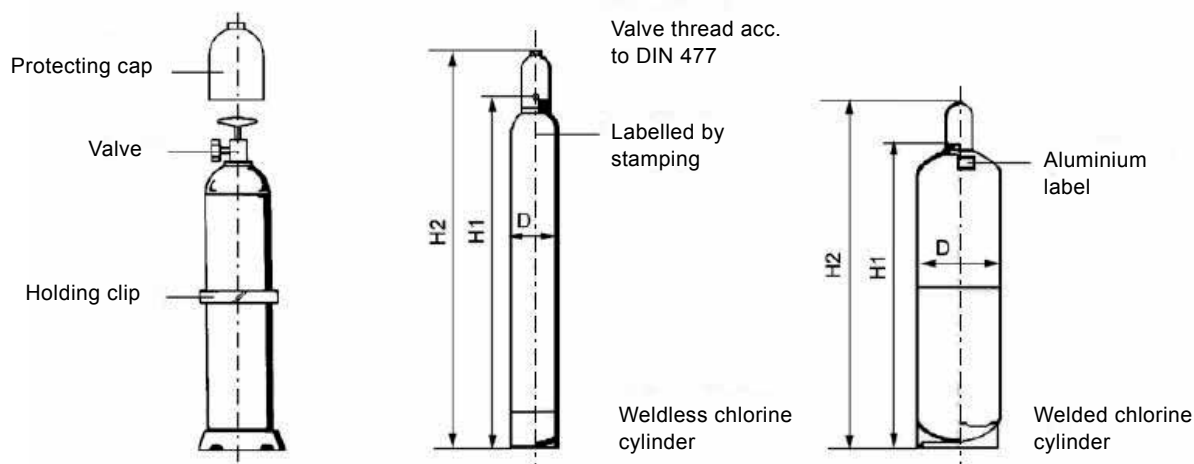
Strictly observe local law and regulations for handling, transport and storage of chlorine.

2.2.6 Pressure vessels and mountings

Chlorine is being offered in two container designs:

- Steel cylinders containing 50 kg or 65 kg, equipped with one valve for
 - the withdrawal of gaseous chlorine from the upright standing cylinder

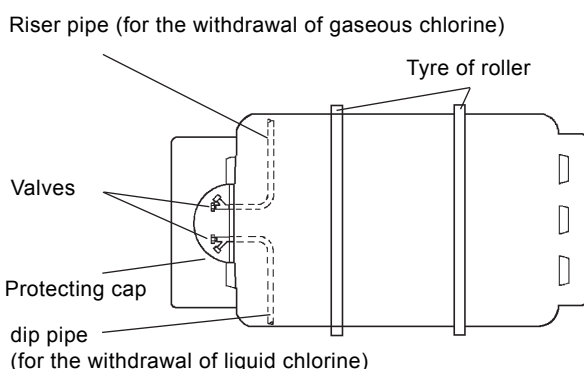
Variant of chlorine cylinder



TM04 0693 0908

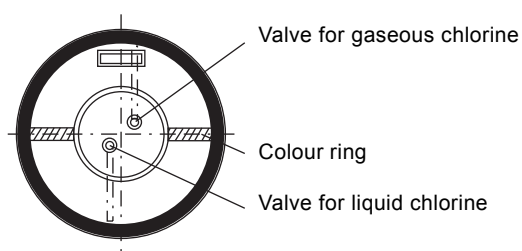
Fig. 3 Variant of chlorine cylinder

- Steel drums containing 500 kg or 1000 kg, equipped with
 - one valve for the withdrawal of gaseous chlorine
 - one valve for the withdrawal of liquid chlorine



TM04 0694 0908

Fig. 4 Chlorine drum



TM04 0695 0908

Fig. 5 Valve position of chlorine drum

Due to safety precautions, chlorine containers are only filled up to 95 % of their capacity.



Warning
Observe safety precautions for chlorine containers

- Kind of gas, weight, owner, producing date and date of the last testing have to be noted clearly on the container. Chlorine containers are marked by grey colour.
- No changes or repair by the user!
- Never open container valves by force. Stuck valve spindels can be loosened by wrapping a shroud with warm water around the valve.
 - Never use an open flame!
 - Never use wrench lengthening!
 - Return containers with stuck valves to the manufacturer.
- Observe safety precautions and the manuals of the manufacturer!

2.2.7 Chlorine extraction

Before the extraction

- The chlorine containers must be stored at least for 8 hours in the container room so that the content can adapt the ambient temperature.



Warning

Chlorine containers must never have a higher temperature than other parts of the plant. Danger of liquefaction and possible chlorine break-out!

- Turn chlorine drums on the support until the dip pipe and the riser pipe are placed vertically (observe the markings on the drum).
- Check tightness.

Connection

- Protect containers from turning over or rolling away!
- Dry the piping and the withdrawal system with dry nitrogen or dry air.
- No foreign matter must get into the plant.
- Mount new gaskets to the connection line.
- Connect the container.
- Slowly open the container valve.

Withdrawal of gaseous chlorine

Caution

At 15 °C about 1 % (DIN19607) of the content can be withdrawn per hour. In case of higher withdrawal danger of malfunction by the formation of ice on containers and piping!

Container size	Withdrawal quantity
50 Kg	500 g/h
65 Kg	650 g/h
500 Kg	5 Kg/h
1000 Kg	10 Kg/h

In case of higher chlorine requirement several chlorine containers of the same temperature have to be connected with header lines.

Withdrawal of liquid chlorine

- Within chlorination plants only possible from chlorine drums.
- Application of an evaporator is necessary.

Caution

Do not completely evacuate the chlorine drums. Danger of withdrawing deposits!

- Withdrawal of gaseous chlorine: Observe residual pressure of ca. 2 bar.
- Withdrawal of liquid chlorine: Observe residual pressure of ca. 4 bar.

After the extraction

- Close container valve.
- Disconnect container from the plant.
- Screw on valve protection nuts.
- Screw on protection cap.

Caution

Immediately close connection lines. No moisture must get into the lines!

2.3 Checking the tightness

Before commissioning check the tightness of the **whole plant**.



Warning

Do not check the tightness until the whole plant is ready for start-up. Danger of chlorine break-out!

2.3.1 Checking the chlorine solution lines and the diaphragm non-return of the injector

- Observe the manual of the injector!

2.3.2 Checking the tightness of the vacuum lines

Vacuum lines are all lines between vacuum regulator and injector.

- Close all container valves
- Close the rate valve
- Open the shut-off valve at the injection unit
- Open motive water valve
- Switch on the booster pump
- Open the rate valve
 - Floater shows gas flow or vacuummeter shows more than - 9 m w. c.: **Leakage in vacuum line!**
- Close rate valve
- Switch off booster pump
- Close motive water valve
- Close the shut-off valve at the injection unit
- Check the vacuum lines and the connection. If necessary, carefully re-tighten them.
- **Check tightness again!**
 - Floater shows no gas flow, vacuummeter shows -9 m w. c. or less: **Vacuum lines are tight.**

More possible reasons for insufficient operating vacuum

Note

- injector layed out too weakly or defective
- injector obstructed
- booster pump layed out too weakly or defective

2.3.3 Checking the tightness of pressure gas lines

Pressure gas lines are all lines from the gas containers to the vacuum regulator.

- If the plant is equipped with a nitrogen rinsing device: Check tightness roughly with nitrogen
- Detailed checking with ammonia

Checking the tightness with nitrogen

- Close all container valves
- Open container connection valves and all shut-off valves up to the gas dosing system
- Open the connection valve of the nitrogen cylinder
- Slowly open the valve of the nitrogen cylinder, until the lines have a pressure of about 10 bar (read at the manometer of the vacuum regulator).



Warning

Maximum nitrogen pressure 16 bar! Danger of damages and gas leakage when being exceeded!

- Apply soap water to all components under pressure
 - Formation of bubbles and/or pressure drop at manometer**Leakage in pressure lines!**
 - **Depressurize the plant!**
 - **Eliminate leakage!**
 - **Check tightness again!**

No formation of bubbles, pressure at manometer does not drop significantly within one hour: **Pressure lines are tight.**

Checking the tightness with ammonia

See chapter 5.2.2 *Checking the tightness with ammonia*.

2.4 Constructional requirements of chlorination plants

All chlorine rooms



Fig. 6 Warning sign DIN 4844

- Warning sign according to DIN 4844 part 1 no. 4.2, to be installed at entrances

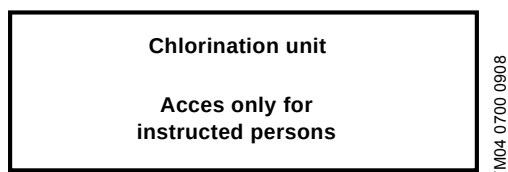


Fig. 7 Supplementary sign

- Supplementary sign according to DIN 4844 part 1 no. 4.5, to be installed at entrances



Fig. 8 Mandatory sign

- Mandatory sign according to DIN 4844 part 1 no. 4.3, to be installed inside the rooms

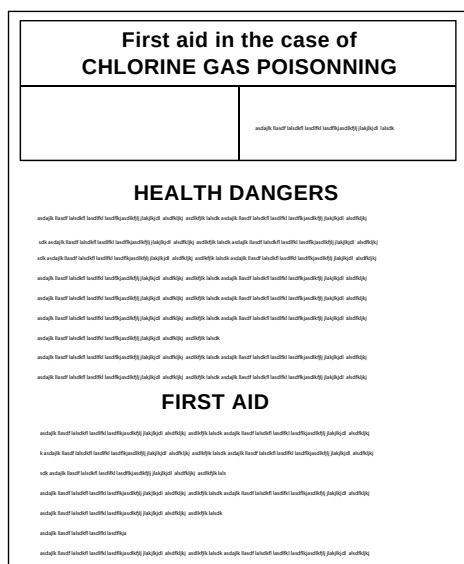


Fig. 9 Instruction sheet for first aid

- Instruction sheet for first aid in case of chlorine gas intoxications, to be installed inside the rooms
- Maximum temperature: 50 °C
 - recommended temperature: 18 - 20 °C
 - recommended minimal temperature: 15 °C
- Overpressure lines of dosing equipment must not end in the open air
- Chlorine rooms must not be dedicated for the permanent stay of people.
- Only chlorine containers and the chlorination plant may be present in the rooms.

Rooms with pressure lines (e. g. storage rooms for chlorine containers)

- Flat, even floor
 - not below ground level
 - not higher than a possible loading ramp

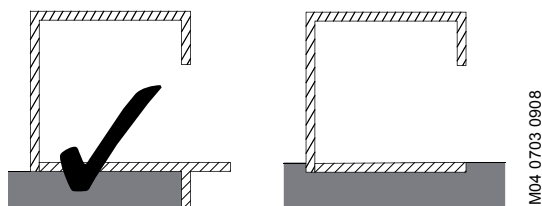


Fig. 10 Regulations for chlorine rooms (1)

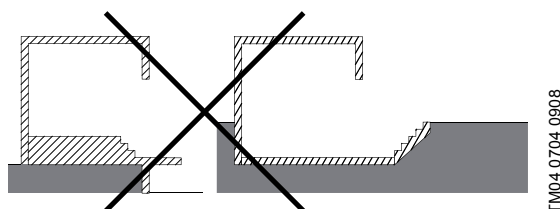


Fig. 11 Regulations for chlorine rooms (2)

- Direct exit to the open air
- Lockable
 - doors must open outwards
 - it must be possible to open the doors without a key from inside the room
- No connection to other rooms
 - separated gastight and fire-resistant from other rooms
- A maximum of two vent holes of max. 20 cm² each
- Water sprinkling system
 - for precipitating escaping chlorine gas
 - operation must be possible by hand from outside the chlorine rooms
 - sufficiently dimensioned run-off with air trap
- Chlorine gas warning system
 - with optical and acoustical alarm
 - coupled to the water sprinkling system
 - coupling must reactivate automatically after having switched off (e. g. for container exchange)
- Chlorine gas must not be able to get into lower-lying rooms, shafts, pits, canals or aspirating holes of ventilation systems.

2.5 Principle function of the components

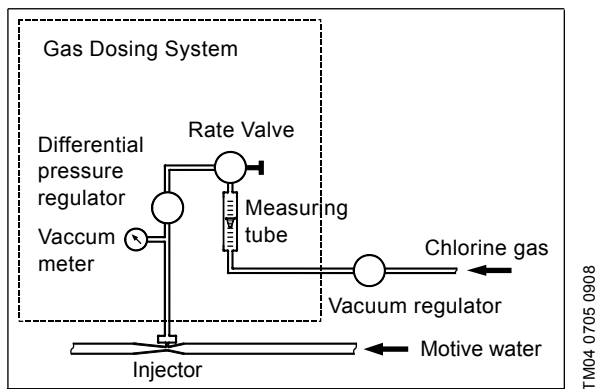


Fig. 12 Components of a chlorinator

2.5.1 Vacuum regulator

- Reduces the gas pressure (overpressure) to subatmospheric pressure (vacuum)

2.5.2 Measuring tube

- Displays the gas flow

2.5.3 Rate valve

- For adjustment of the required gas flow
 - manually or automatically

2.5.4 Differential pressure regulator (Option: VGA-117)

- Regulates the difference of the pressures before and after the rate valve to a constant value
 - Adjusted dosing flow stays constant even when the injector vacuum varies.

2.5.5 Vacuummeter (Option: VGA-117)

- Displays the injector vacuum

2.5.6 Injector

- Creates the vacuum necessary for operating the pant
- Mixes the chlorine gas with the water

2.6 List of valid laws and regulations

Laws and Regulations

BGBI. I 1975 S. 729	Verordnung über Arbeitsstätten (Arbeitsstättenverordnung - ArbStättV) Regulations concerning places of work (ArbStättV)
BGBI I 1980 S. 173, 184	Verordnung über Druckbehälter, Druckgasbehälter und Füllanlagen (Druckbehälterverordnung - DruckbehV) mit technischen Regeln Druckbehälter (TRB) mit technischen Regeln Druckgase (TRG) Regulations concerning pressure containers, gas cylinders and filling systems (pressure containers - DruckbehV) with technical rules for pressure containers (TRB) with technical rules for pressure gases (TRG)
BGBI I 1986 S. 1470	Gefahrstoffverordnung (GefStoffV) Regulations for hazardous materials (GefStoffV)
BGBI I 1975 S. 2494	Verordnung über gefährliche Arbeitsstoffe (Arbeitsstoffverordnung - ArbStoffV) Regulations for hazardous work materials (ArbStoffV)
GUV 0.1	Unfallverhütungsvorschrift "Allgemeine Vorschriften" Regulations for accident prevention "General Regulations"
GUV-V D5	Unfallverhütungsvorschrift "Chlorung von Wasser" Regulations for accident prevention "Chlorination of water"
GUV 49.1	Prüfliste zur Unfallverhütungsvorschrift "Chlorung von Wasser" Checklist for Regulations for accident prevention "Chlorination of water"
GUV 0.3	Unfallverhütungsvorschrift "Erste Hilfe" Regulations for accident prevention "First-Aid"
GUV 20.5	Merkblatt "Anleitung zur ersten Hilfe bei Unfällen" Leaflet "Instructions for First-Aid in case of accidents"
GUV 20.6	Merkblatt "Verbandzeug für die erste Hilfe bei Unfällen" Leaflet "Dressing material for First-Aid in case of accidents"
GUV 20.14	Atenschutzmerkblatt Leaflet for the protection of the air
GUV 29.6	Merkblatt "Über den Umgang mit ätzenden Stoffen" Leaflet "Handling of caustic materials"
GUV 60.3G26	Grundsatz für arbeitsmedizinische Vorsorgeuntersuchungen "Träger von Atemschutzgeräten für Arbeit und Rettung" G 26 Principle for preventive industrial medicine checkup "Wearers of respiratory equipment for work and rescue" G 26
GUV 2.6	Unfallverhütungsvorschrift "Druckbehälter" Regulations for accident prevention "Pressure containers"
GUV 2.10	Unfallverhütungsvorschrift "Elektrische Anlagen und Betriebsmittel" Regulations for accident prevention "Electrical installations and resources"
GUV 9.9	Unfallverhütungsvorschrift "Gase" Regulations for accident prevention "Gases"
ZH1/230	Merkblatt "Chlor" Leaflet "Chlorine"
CEFIC	Unfallmerkblatt für den Straßentransport "Chlor" Klasse 2, Ziffer 3 at UN 2201 Accident leaflet for the road transport "Chlorine" Class 2, Number 3 at UN 2201
BGBI I 1985, S. 1550	Verordnung über die Beförderung gefährlicher Güter auf der Straße - Gefahrgut Vstr/GGVS - Klasse 2, Ziffer 3 at Regulations for the transport of dangerous goods on the road - Gefahrgut Vstr/GGVS - Class 2, Number 3 at
GGVE	Gefahrgutverordnung Eisenbahn, Klasse 2, Ziffer 3 at) Regulations for dangerous goods on trains, Class 2, Number 3 at)
RID	Internationale Verordnung für die Beförderung gefährlicher Güter mit der Eisenbahn - Klasse 2, Ziffer 3 at International regulations for the transport of dangerous goods by train - Class 2, Number 3 at
BGBI I 1977, S. 1119	Verordnung über die Beförderung gefährlicher Güter auf dem Rhein (ADNR) Regulations for the transport of dangerous goods on the Rhine (ADNR)
DVGW-Rules	
W203	Begriffe der Chlorung Concept of chlorination
W645-1	Überwachungs-, Mess-, Steuer- und Regeleinrichtungen in Wasserversorgungsanlagen Devices for monitoring, measuring, control and regulation in water supply plants
W291	Desinfektion von Wasserversorgungsanlagen Disinfection of water supply plants
W623	Dosieranlagen für Desinfektions- bzw. Oxidationsmittel - Dosieranlagen für Chlor Dosing units for disinfection or oxidation with chlorine
W640	Überwachungs-, Meß-, Steuer- und Regeleinrichtungen in Wasserwerken Systems for monitoring, measurement, control and regulation in waterworks DVGW-Merkblatt Arbeitshilfe zur Erstellung einer örtlichen Betriebsanweisung für Chlorungsanlagen unter Verwendung von Chlorgas DVGW leaflet Aid for the creation of a local manual für chlorination systems using chlorine gas

Standards

DIN 19606	Chlorgasdosieranlagen zur Wasseraufbereitung Chlorine gas dosing systems for water treatment
DIN 19607	Chlor zur Wasseraufbereitung Chlorine for water treatment
DIN EN 937	Chlor zur Aufbereitung von Wasser für den menschlichen Gebrauch Chlorine for the treatment of water for the human use
DIN 19643	Aufbereitung von Schwimm- und Badewasser Treatment of swimming pool and bathing water
DIN 3179, Teil 1,2	Einteilung der Atemgeräte, Übersicht Division of the respiratory equipment, overview
DIN 4102, Teil 2	Brandverhalten von Baustoffen und Bauteilen Behaviour in fire of building materials and parts
DIN 477, Teil 1	Gasflaschenventile; Bauformen, Baumaße, Anschlüsse, Gewinde Gas cylinder valves; forms, measurements, connections, threads

2.7 Recommended diameter

2.7.1 Between vacuum regulator and dosing regulator

Length of the vacuum line in (m)	Dosing quantity (g/h)						
	500	1000	2000	4000	6000	8000	10000
0	DN 8	DN 8	DN 8	DN 10	DN 10	DN 10	DN 15
10	DN 8	DN 8	DN 8	DN 10	DN 10	DN 15	DN 15
20	DN 8	DN 8	DN 10	DN 15	DN 15	DN 15	DN 20
30	DN 8	DN 8	DN 10	DN 15	DN 15	DN 20	DN 20
40	DN 8	DN 8	DN 15	DN 15	DN 15	DN 20	DN 20
50	DN 8	DN 10	DN 15	DN 15	DN 15	DN 20	DN 20
75	DN 8	DN 10	DN 15	DN 15	DN 20	DN 20	DN 25
100	DN 8	DN 10	DN 15	DN 20	DN 20	DN 25	DN 25

Grundfos Alldos company standard calculated with pressure drop $p=12.5$ mbar

The indications in this table result from pressure loss observations. They do not take into consideration the possible influence of length and diameter of lines on the operational reliability of the system.

2.7.2 Between dosing regulator and injector

Length of the vacuum line in (m)	Dosing quantity (g/h)						
	500	1000	2000	4000	6000	8000	10000
0	DN 8	DN 8	DN 8	DN 8	DN 10	DN 10	DN 15
10	DN 8	DN 8	DN 8	DN 8	DN 10	DN 10	DN 15
20	DN 8	DN 8	DN 8	DN 10	DN 10	DN 15	DN 15
30	DN 8	DN 8	DN 8	DN 10	DN 10	DN 15	DN 15
40	DN 8	DN 8	DN 8	DN 10	DN 15	DN 15	DN 15
50	DN 8	DN 8	DN 10	DN 15	DN 15	DN 15	DN 15
75	DN 8	DN 8	DN 10	DN 15	DN 15	DN 15	DN 20
100	DN 8	DN 8	DN 10	DN 15	DN 15	DN 15	DN 20

Grundfos Alldos company standard calculated with pressure drop $p=50$ mbar

The indications in this table result from pressure loss observations. They do not take into consideration the possible influence of length and diameter of lines on the operational reliability of the system.

3. Technical Data

3.1 Type key dosing regulator vaccuperm VGA-117

Example: Dosing regulator Type key VGA-117-1000/1-M0X

Code	Example	VGA-117	-1000	/5	-M	O	, N
	Vaccuperm Gas Advanced = VGA						
Dosing flow							
500	25-500 g/h						
1000	50-1000 g/h						
2000	100-2000 g/h						
4000	200-4000 g/h						
6000	300-6000 g/h						
8000	400-8000 g/h						
10000	500-10000 g/h						
Connection input and output							
1	PE hose 8/11						
2	PE hose 10/14						
3	PVC pipe DN15 (d 20)						
Adjustment							
M	Manual						
A	Automatic, 230-240V/50-60Hz						
B	Automatic, 110-115V/50-60Hz						
Control							
0	Without						
1	4-20 mA						
2	Direct						
Differential pressure regulator							
Y	Yes, with vaccummeter						
N	No						

3.2 General data

Accuracy	± 4 % of upper limit
Control range	1 : 20
Permissible medium	Cl ₂
Permissible vacuum regulator	VGA-111, VGA-146: up to 4000g/h
Flow meter	according to the floater principle, length of the measuring tube 190 mm
Weight	without servomotor: 2.6 kg with servomotor: 4.2 kg

3.2.1 Vacuum connection (from the vacuum regulator / to the injector)

PE hose 8/11 or 10/14
PE hose 8/11 or 10/14
PVC pipe DN15 (d20)

3.2.2 Accessories (not including)

Installation material: Hoses of different lengths
Holding plate for wall fixing at change of container
Test medium for leak search

3.2.3 Dosing Flow

50 - 1000 g/h	2 - 50 lbs/day
100 - 2000 g/h	5 - 100 lbs/day
200 - 4000 g/h	10 - 210 lbs/day
300 - 6000 g/h	15 - 310 lbs/day
400 - 8000 g/h	20 - 420 lbs/day
500 - 10000 g/h	30 - 520 lbs/day

3.2.4 Accessories (not including)

Installation material: Hoses of different lengths
Holding plate for wall fixing at change of container
Test medium for leak search

3.3 Dimensioned drawings

3.3.1 Dosing unit VGA-117

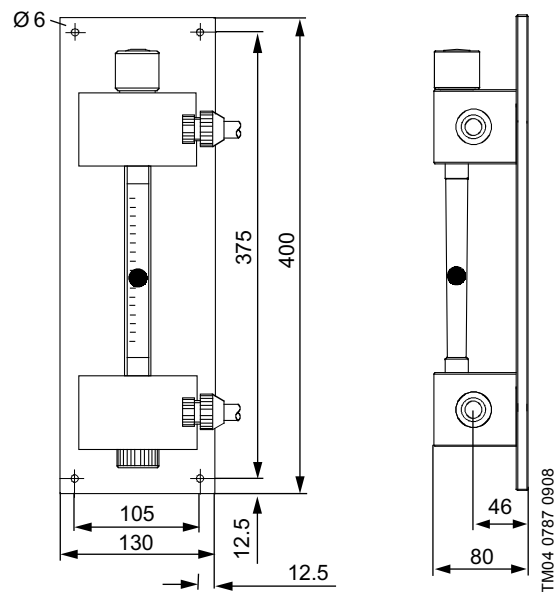


Fig. 13 Dimensional drawing dosing unit VGA-117

3.3.2 Dosing regulator with servomotor

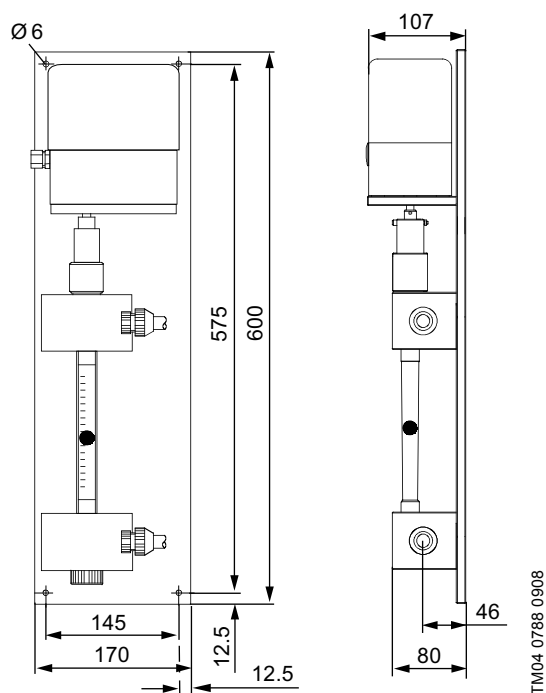


Fig. 14 Dimensional drawing dosing regulator VGA-117 with servomotor

Item	Description
1	SR, limit switch cam min. SL, limit switch cam max.
2	Supply voltage input
3	Potentiometer input/output
4	Earth terminal
Mains voltage	220 - 240 V $\pm 10\%$, 50/60 Hz $\pm 5\%$, 100 % on time (standard) 110 - 120 V $\pm 10\%$, 50/60 Hz $\pm 5\%$ 24 V AC $\pm 10\%$, 50/60 Hz $\pm 5\%$
Rated current	30 mA (at 230 V), 60 mA (at 115 V), 240 mA (at 24 V)
Limit switch - load capacity	2 A at 250 V
Power consumption	Approx. 2 VA
Control variant	Reversing potentiometer 1 k Ω Open/close signal
Actuating time	90 s for 270° at 50 Hz 75 s for 270° at 60 Hz
Adjustment angle	Max. 270°
Degree of protection	IP 65
Permitted ambient temperature	-15 to 60

3.4 Electrical data

3.4.1 Servomotor with reversing potentiometer

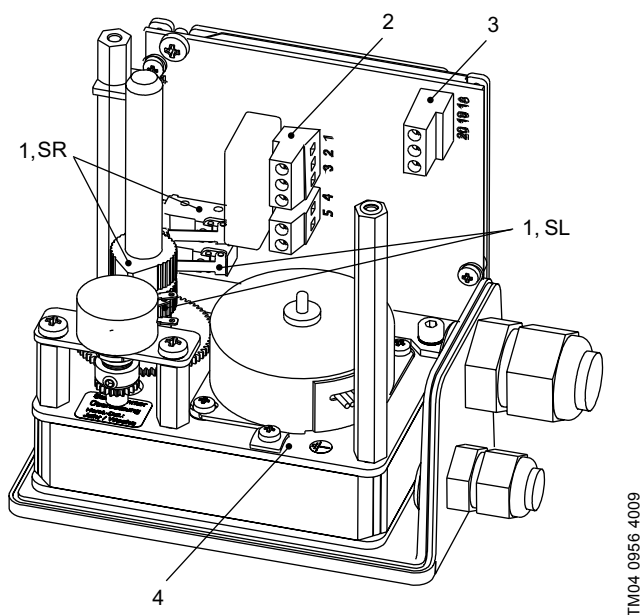
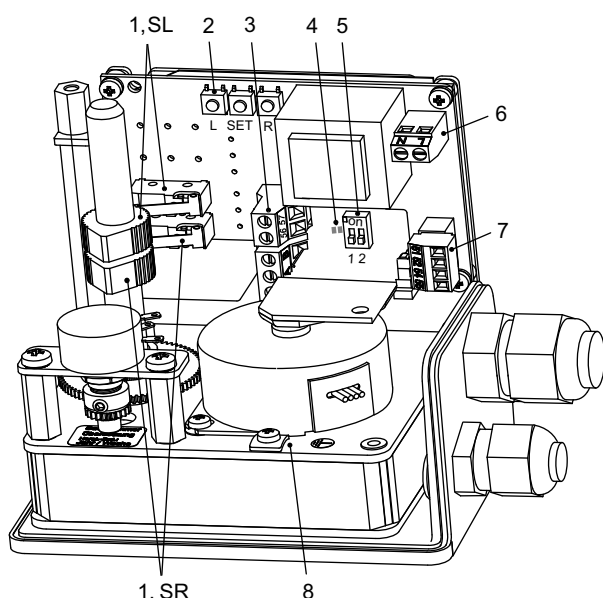


Fig. 15 Servomotor with reversing potentiometer

3.4.2 Servomotor with analogue control (4-20 mA/0-20 mA)



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Fig. 16 Servomotor with analogue control

Item	Description
1	SR, limit switch cam min. SL, limit switch cam max.
2	Programming buttons
3	Alarm output
4	LEDs
5	DIP switch
6	Mains connection terminals
7	Analogue signal connection terminals
8	Earth terminal
Mains voltage	220 - 240 V $\pm 10\%$, 50/60 Hz $\pm 5\%$, 100 % on time (standard) 110 - 120 V $\pm 10\%$, 50/60 Hz $\pm 5\%$ 24 V AC $\pm 10\%$, 50/60 Hz $\pm 5\%$
Rated current	30 mA (at 230 V), 60 mA (at 115 V), 240 mA (at 24 V)
Limit switch - load capacity	2 A at 250 V
Power consumption	Approx. 2 VA
Control variant	Analogue control Input/Output
Signal input/output	4 - 20 mA (standard setting) 0 - 20 mA (switchable) 4 (0) - 20 mA (DC) Load 250 Ω
Setpoint input	Overload protection 25 mA Protection against reverse polarity to -25 mA ESD protection, input filter Resolution: 10 bits 4 (0) - 20 mA
Actual value output	Max. load 500 Ω Resolution: 10 bits
Carrying capacity of signal output	Output current: 4 - 20 mA Minimum voltage for open circuit 10 V
Potentiometer	1000 Ω

Mains voltage	220 - 240 V $\pm 10\%$, 50/60 Hz $\pm 5\%$, 100 % on time (standard) 110 - 120 V $\pm 10\%$, 50/60 Hz $\pm 5\%$ 24 V AC $\pm 10\%$, 50/60 Hz $\pm 5\%$
Actuating time	90 s for 270° at 50 Hz 75 s for 270° at 60 Hz
Adjustment angle	Max. 270°
Degree of protection	IP 65
Permitted ambient temperature	0 to 60

4. Installation

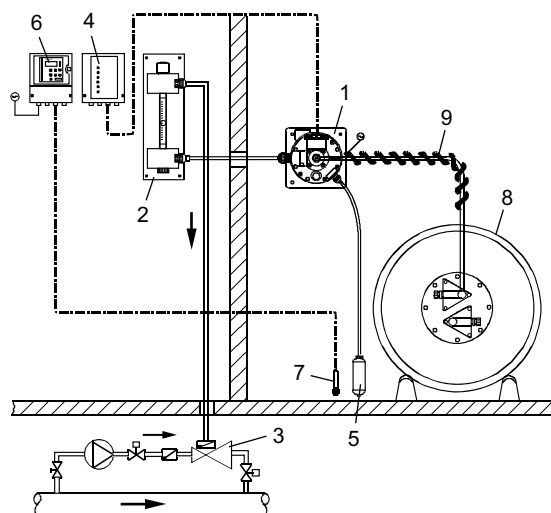
4.1 Transport and storage

- Handle with care, do not throw!
- Dry and cool storage place.

4.2 Unpacking

- Observe when unpacking:
 - No humidity should get into gas-leading parts!
 - No foreign matter should get into gas-leading parts!
- Mount as soon as possible after unpacking.

4.3 Typical installation



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Fig. 17 Dosing and vacuum regulator

Item	Description
1	Vacuum regulator for wall mounting
2	Dosing regulator
3	Injector
5	Remote display for container empty indication
6	Adsorption device
8	Gas warning device
11	Gas sensor
12	Gas drum
13	Heater strip

4.4 Mounting



Warning

Observe the information, see chapter 2. Handling chlorine

- Mount the dosing unit to the wall.

5. Commissioning

5.1 Preparations for commissioning

5.1.1 Vacuum connections

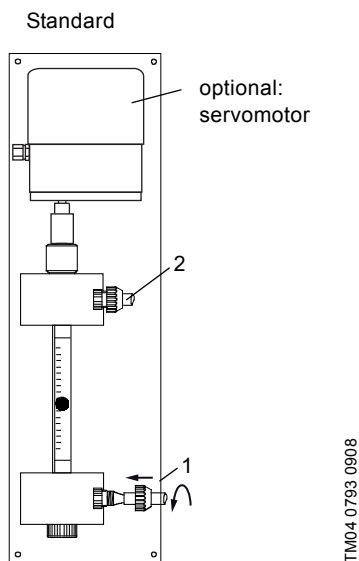


Fig. 18 Vacuum connections, standard

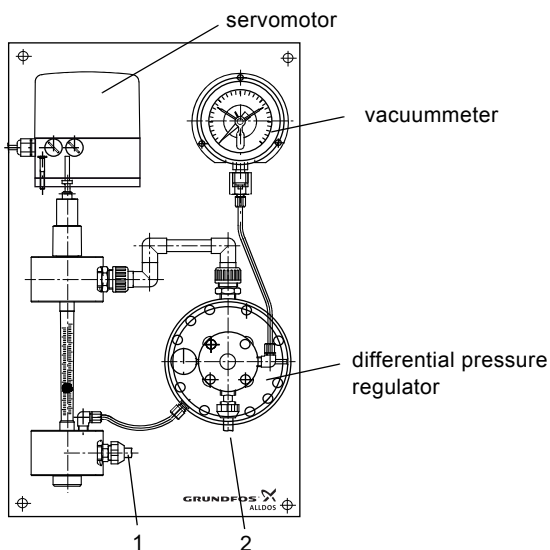


Fig. 19 Vacuum connections, none standard, with optional accessories



Warning

**Before connecting, ensure that the valves of all gas containers are closed.
Only use the intended line types!**

Item	Description
1	Vacuum line from the vacuum regulator
2	Vacuum line to the injector

Only tighten the union nuts of the vacuum connections by hand. Do not use any tools!

Caution

Danger of damages!

Maximum length of the vacuum line, see chapter 2.7 Recommended diameter

- Connect the vacuum line from the vacuum regulator (1).
- Connect the vacuum line to the injector (2).

5.1.2 Electrical connections dosing regulator - with servomotor

Warning

Disconnect from mains before removing the cover!

Observe local safety precautions!

Protect cable connections and plugs against corrosion and moisture.

Close unused screwings with suitable caps!

Before connecting the mains cable:

Check if the mains voltage indicated on the type plate corresponds to the local mains voltage. Wrong voltage can lead to damages of the product!

Caution

**To ensure electromagnetic compatibility (EMC):
Signal lines must be shielded.**

Caution

**The shield must be connected to PE or ground on the side of the counterpart.
Signal lines and mains lines must be guided in different cable channels.**

- Establish the electrical connections according to the following terminal diagrams.

Note **Depends on the optional components selected:
Direct mains connection (by customer) or via the terminal box.**

5.1.3 Servomotor with analogue control (optional)

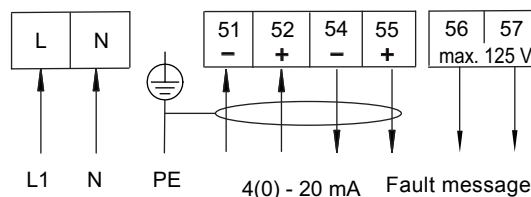


Fig. 20 Circuit diagram for a servomotor with analogue control

5.1.4 Servomotor with analogue control (optional), connected, for example, to a Conex® DIA-2Q

- Connect the terminals to the corresponding terminals on the external control device as shown in the connection diagram.

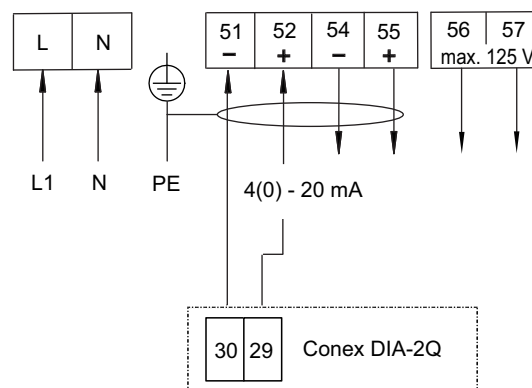
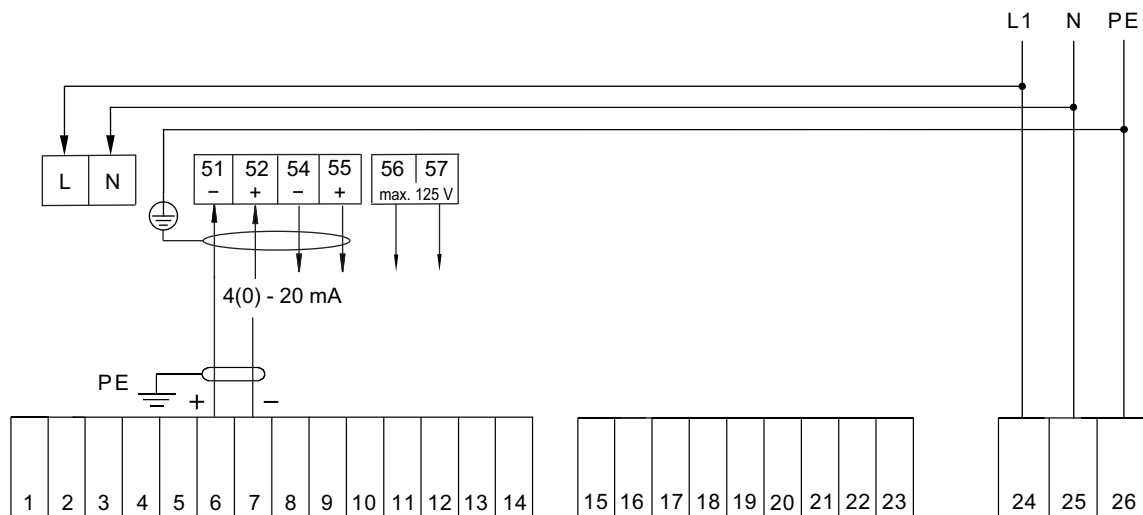


Fig. 21 Terminal connection diagram for Conex® DIA-2Q

5.1.5 Servomotor with analogue control (optional), connected, for example, to a Conex® DIS-2Q (microprocessor-based control device)

- Connect the terminals to the corresponding terminals on the external control device as shown in the connection diagram.



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Fig. 22 Terminal connection diagram for Conex® DIS-2Q on a servomotor with analogue control

Item	Description
L	Phase
N	Neutral
PE	PE (earth)
51	Signal input (-)
52	Signal input (+)
54	Signal output (-)
55	Signal output (+)
56	Collective error message
57	

Note

See Section 5.2.4 for warnings and notes relating to electrical connection!

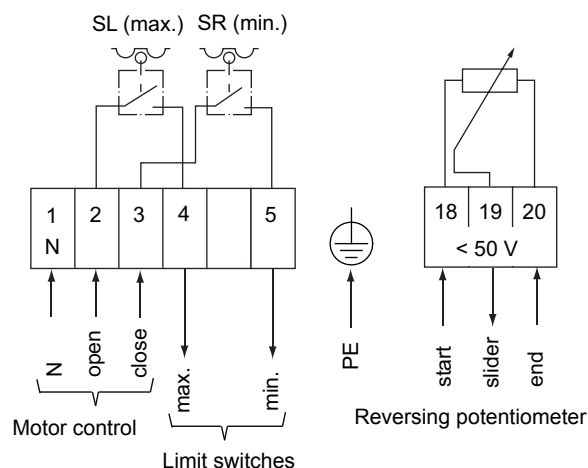
- Unscrew the hood screws and remove the hood.
- Establish the electrical connections in accordance with the circuit diagram; see Fig. Terminal connection diagram for Conex® DIA-2Q.
 - Connect terminals 51 and 52 and, if required, terminals 54 and 55, in accordance with the circuit diagram.
 - Connect terminals 56 and 57 in accordance with requirements; voltage 125 V max.
 - Use separate, shielded lines with a minimum cross-section of 0.5 mm² and a maximum length of 1000 m for low voltages (setpoint/actual value signal lines).
 - The screen must be connected on one side to the housing earth (earth terminal).



Warning
The internal wiring of the drive must never be modified.

- Close the hood again and tighten the hood screws.

Servomotor - with reversing potentiometer (optional)



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Fig. 23 Terminal connection diagram for servomotor with potentiometer

Item	Description
1 N	Neutral
2	Phase
3	Phase
4	Mains voltage output (position "final position max.")
5	Mains voltage output (position "final position min.")
PE	PE (earth)
18	Start
19	Slider
20	End

- Connect the terminals to the corresponding terminals on the external control device (Conex® DIS-2Q in the example) as shown in the connection diagram.

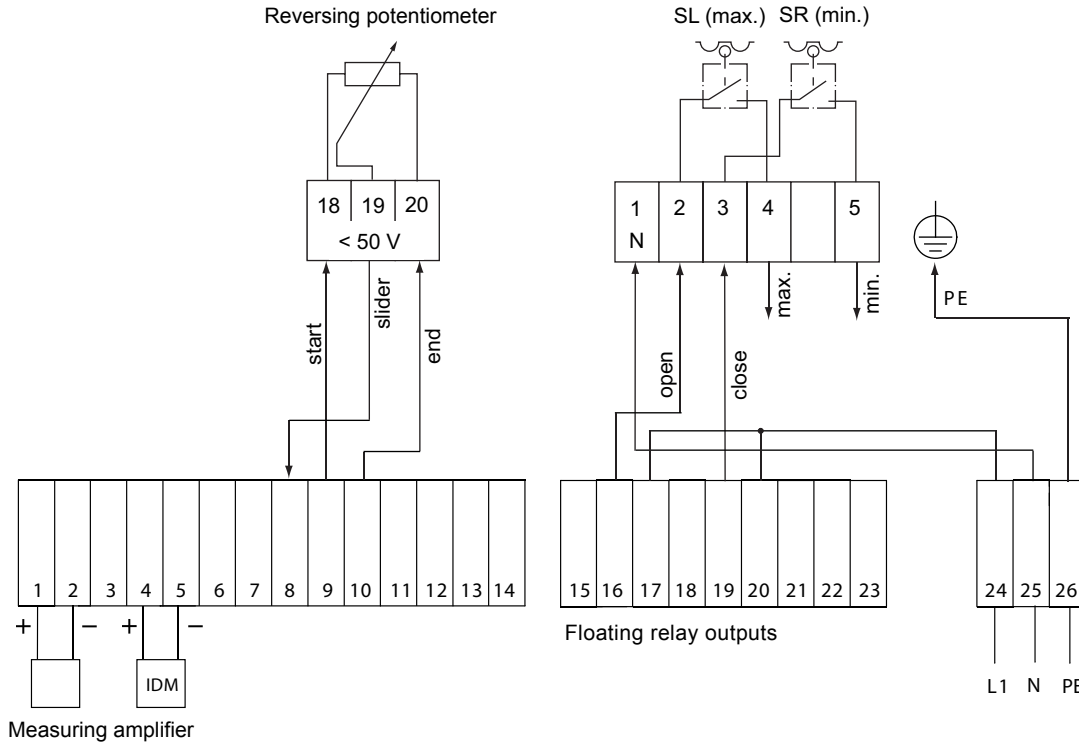


Fig. 24 Terminal connection diagram for Conex® DIS-2Q on a servomotor with potentiometer

Electrical connection

Note See Section 5.2.4 for warnings and notes relating to electrical connection!

1. Unscrew the hood screws and remove the hood.
2. Establish the electrical connections in accordance with the circuit diagram; see Fig. Terminal connection diagram for servomotor with potentiometer.
 - Connect terminals 18, 19 and 20 in accordance with requirements; voltage < 50 V.
3. Determine the direction of rotation and, if necessary, correct it.

Determining the direction of rotation

The internal wiring means that the direction of rotation (looking through the drive towards the output shaft) and limit switches are assigned as follows:

- If the mains voltage is present at terminal 2, the output shaft rotates anticlockwise (in the direction "open/max.").
 - This direction of rotation is restricted by the upper switch (max.). If the switch is activated, the mains voltage is present at terminal 4.
- If the mains voltage is present at terminal 3, the output shaft rotates clockwise (in the direction "close/min.").
 - This direction of rotation is restricted by the lower switch (min.). If the switch is activated, the mains voltage is present at terminal 5.
- If the drive rotates in the opposite direction to the control commands, swap over the connections of terminals 2 and 3.



Warning
The internal wiring of the drive must never be modified.

Close the hood again and tighten the hood screws.

5.2 Checks before commissioning

Check tightness of the total plant before start-up.



Warning
Check the tightness not until the total plant is ready for start-up.
Danger of gas leakage!

5.2.1 Checking the gas solution lines and the non-return diaphragm of the injector

- Observe the manual of the injector!

5.2.2 Checking the tightness with ammonia



Warning
Liquid ammonia must not come in contact with parts of the plant!
Danger of leakages by corrosion!



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Fig. 25 No liquid ammonia

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5.2.3 Checking the pressure connections (after changing gas cylinder)

- Open the cylinder valve and quickly close it again.
- Slowly pass the open ammonia bottle along gas-leading parts.
 - Slightly press the bottle in a pumping manner, allowing the ammonia mist to rise up.
 - Formation of white mist: Leakage at the pressure connection!

• Depressurize the plant!

• Eliminate leakage!

• Check tightness again

Formation of white mist --> pressure connection is not tight

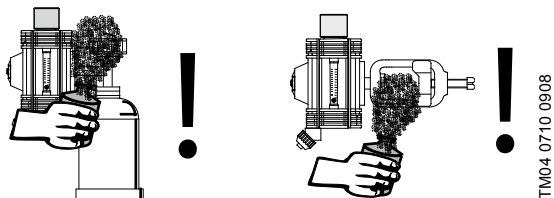


Fig. 26 Ammonia + chlorine gas --> formation of white mist

No formation of white mist --> pressure connection is tight

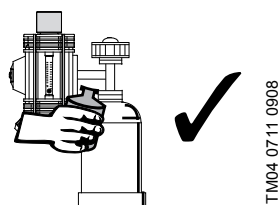


Fig. 27 No chlorine gas --> no formation of white mist

5.2.4 Checking the inlet valve

- Remove the union nut at the overpressure connection and take off the overpressure line.
- Close the rate valve.
- Open the cylinder valve.
- Slowly pass the open ammonia bottle along the overpressure connection.
 - Slightly press the bottle in a pumping manner, allowing the ammonia mist to rise up.
 - Formation of white mist: **The inlet valve is not tight!**
- **Depressurize the plant!**
- **Check the inlet valve and repair it!**
- **Check tightness again**

Formation of white mist --> the inlet valve is not tight!

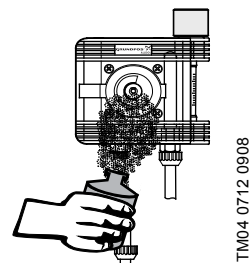


Fig. 28 Ammonia + chlorine gas --> formation of white mist

No formation of white mist --> the inlet valve is tight!

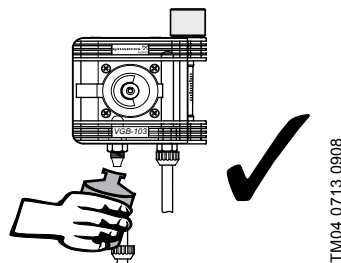


Fig. 29 No chlorine gas --> no formation of white mist

6. Operation

6.1 Description of the device

6.1.1 Dosing regulator

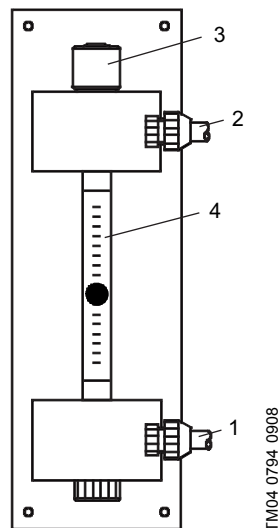


Fig. 30 Schema dosing regulator VGA-117

Item	Description
1	Connection for the vacuum line from the vacuum regulator
2	Connection for the vacuum line to the injector
3	Rate valve: for manual adjustment of the desired gas flow.
4	Measuring tube: It displays the actual gas flow

Caution The rate valve is not a perfectly closing shut-off valve!

6.1.2 Dosing regulator with servo motor

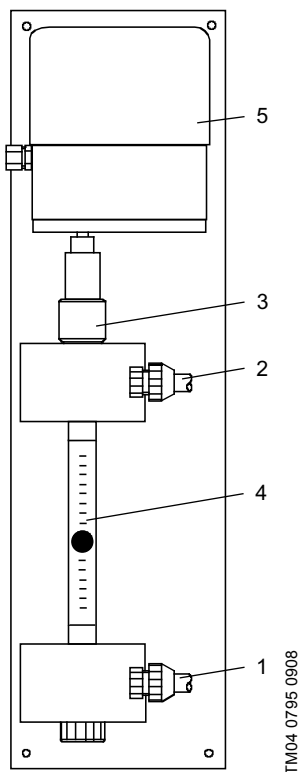


Fig. 31 Schema dosing unit VGA-117 with servomotor

Item	Description
1	Connection for the vacuum line from the vacuum regulator
2	Connection for the vacuum line to the injector
3, 5	Rate valve with servomotor: For automatic adjustment of the desired gas flow
4	Measuring tube: It displays the actual gas flow

Caution *The rate valve is not a perfectly closing shut-off valve!*

6.2 Adjusting the Dosing Flow

6.2.1 Increasing the dosing flow

- Turn the adjustment knob of the rate valve **slowly** to the right.

Caution *Do not screw out the adjustment knob. Do not continue turning the knob when maximum dosing flow is reached.*

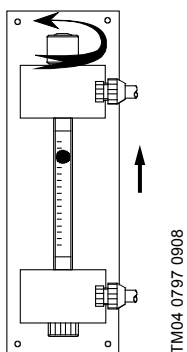


Fig. 32 Increasing the dosing flow

6.2.2 Decreasing the dosing flow

- Turn the adjustment knob of the rate valve slowly to the left.

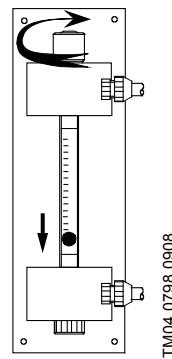


Fig. 33 Decreasing the dosing flow

6.3 Reading the dosing flow

Note

The scale of the measuring tube is adjusted to a gas temperature of 20 °C. In case of extremely different temperatures the dosing flow may differ from the value being displayed.

- Read the dosing flow at the upper edge of the ball.

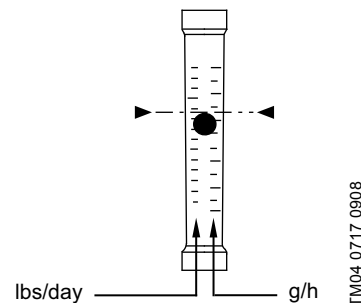


Fig. 34 Reading the dosing flow

6.4 Change between automatic and manual adjustment of the dosing flow

Automatic adjustment of the dosing flow

In case of automatic adjustment of the dosing flow, no adjustment can be made at the dosing unit. All adjustment has to be made at the control device.

Observe the manual of the control device!

6.4.1 Switching from automatic to manual adjustment of the dosing flow (only devices with servomotor)

- Switch off automatic control.
- Push the coupling (1) upwards.
- Move the retention pin (2) through the coupling, until it cannot move down any longer.
 - Now the dosing flow can be adjusted by hand.
- The servomotor is disconnected from the rate valve.

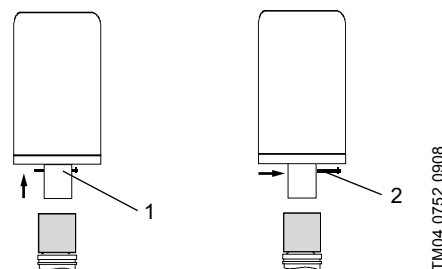


Fig. 35 Disconnect to servomotor

6.4.2 Switching from manual to automatic adjustment of the dosing flow (only devices with servomotor)

- Move back the retention pin (2).
 - Now the coupling (1) falls down.
- Turn the adjustment knob (3), until the coupling snaps in.
 - Now the servomotor is connected to the rate valve.

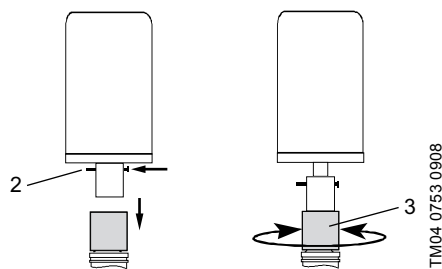


Fig. 36 Connect to servomotor

6.5 Switching on

- Close the rate valve.
- Open the shut-off valve at the injection unit.
- Open the motive water valves.
- Switch on the motive water of the injector.
- Open the valve of the gas cylinder.
- Slowly open the rate valve until the ball in the measuring tube shows the desired gas flow.

6.6 Switching off

6.6.1 Emergency Stop

In case of gas break-out



Warning

Immediately leave the room in case of gas break-out and put on safety equipment!

Then start counter-measures!

- Immediately leave the room!
- Start counter-measures according to local safety precautions.

In case of small leakages

- Put on safety equipment!
- Immediately close the cylinder valve.
- Let the system run until all parts are evacuated.
- Switch off the system as described in the following



Warning

Repairs of components of the system only by authorized personnel!

6.6.2 Switching off while the system is running

- Close the cylinder valve.
- Let the system run until the measuring tube shows no more gas flow.
- Close the rate valve.
- Switch off the motive water of the injector.
- Close the motive water valves.
- Close the shut-off valve at the injection unit.

6.7 Possible faults

Fault	Cause	Correction
Desired dosing flow is not reached.	Insufficient injector vacuum.	Check the motive water pump, repair it if necessary. Observe the manual of the injector.
		Check the motive water pump, repair if necessary.
		Check the dirt trap before the injector, clean it if necessary.
	Leakage in vacuum line between dosing unit and injector.	Eliminate the leakage.
	Vacuum line between dosing unit and injector is too long.	Observe table with suggested line lengths and diameters in chapter 2.7 <i>Recommended diameter</i> .
	Vacuum line is dirty.	Replace the vacuum line.
	Servomotor does not reach the 100 %-position.	Adjust the servomotor or check the settings of the control device.
	Filter at the pressure connection of the dosing unit is soiled.	Call the service
	The valve of the gas container is closed.	Open the valve of the gas container.
	The gas container is empty.	Replace the empty gas container by a new one.
	Wrong adjusting spindle (diameter too low)	Call the service
Despite correctly displayed gas flow the desired gas solution concentration cannot be reached.	Leakage in vacuum line between vacuum regulator and dosing unit.	Eliminate the leakage.
The ball in the measuring tube is caught.	Measuring tube and/or ball are soiled.	Call the service
Water in the measuring tube.	Non-return diaphragm of the injector is defective.	Call the service
Servomotor does not run.	Servomotor is switched to manual operation.	Switch servomotor to automatic operation.
Servomotor does not reach the whole desired range.	The servomotor or the control device is adjusted wrongly.	Call the service
Servomotor runs in wrong direction.		Call the service
Despite constant operating conditions the vacuum varies	Spring in differential pressure regulator is soiled	Call the service
	Spring in differential pressure regulator is damaged	Call the service
Despite constant operating conditions the dosing flow varies	Diaphragm in differential regulator is soiled	Call the service
	Diaphragm in differential regulator is damaged	Call the service

7. Maintenance

Rates for cleaning and maintenance

- at least every 12 months
- in case of malfunction



Warning

Switch off the whole plant before doing cleaning and maintenance work!

Danger of gas break-out!



Warning

Before re-starting, check the tightness!

Danger of gas break-out!

8. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use appropriate waste collection services.
2. If this is not possible, contact the nearest Grundfos or Grundfos Alldos company or service workshop.

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